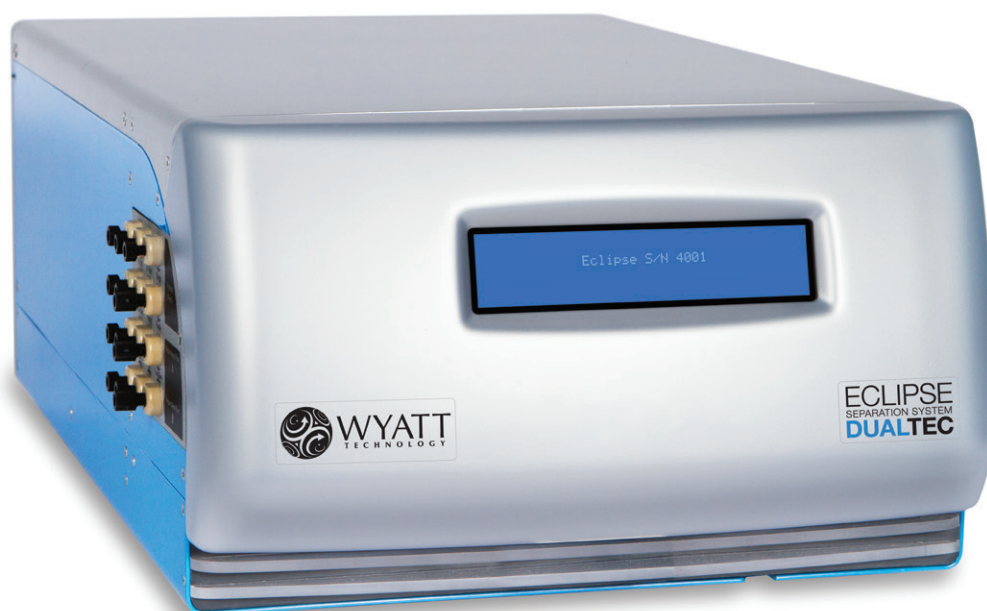


Eclipse

DualTec & AF4

Premier Flow FFF technology for versatile separations



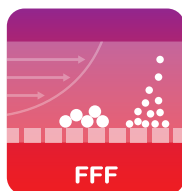
Eclipse®

Separate and characterize nanoparticles, proteins and polymers

Separate & Conquer

Asymmetric-Flow Field-Flow Fractionation (AF4) is an advanced technique for achieving analytical and semi-preparative separations, applicable to a wide range of analytes including:

- Nanoparticles
- Proteins
- Polymers
- Complex samples



Covering an extraordinary range of sizes from 1 nm to 10 µm, AF4 offers programmable separation power yet negligible shear. It does not disrupt delicate macromolecular assemblies or loose protein aggregates. Even the most complex macromolecular solutions and colloidal particle suspensions can be analyzed utilizing multiple online detection methods after separation by AF4, from light scattering to ICP-MS.

Hollow-Fiber Flow Field-Flow Fractionation (HF5), closely related to AF4, performs low-volume analytical separations in convenient disposable cartridges with nanograms of analyte.

	Eclipse DualTec	Eclipse AF4
Description	Most versatile separations	Specialized separations
Injection Method	Tip	Center-Downstream
Dual-channel Switching	Any two of SEC, AF4, HF5	No
Metal-free Sample Path	For ICP-MS	No
Temperature-controlled Separations	4 °C to 90 °C with a temperature regulation chamber	
Channel Options		
Analytical AF4	Long and Short Channels 1 to 100 µg injections	
HF5 Disposable Hollow Fiber	pg to low µg injections	No
Semi-preparative	No	mg separations
Frit-inlet	No	For aggregation-prone samples

Meet the Family

The Eclipse family of flow field-flow fractionation controllers and separation channels offers multiple options to meet the most demanding separation needs.

Eclipse DualTec AF4 controller, supports HF5 and channel switching

Eclipse AF4 AF4 controller, supports semi-preparative and frit-inlet channels

Analytical Separation Channels Two sizes to optimize separation power and sample quantity in the microgram range

HF5 Cartridges Disposable channels for nanogram separations

Semi-preparative Channel Fractionate milligrams of sample

Frit-inlet Channel Suitable for aggregation-prone samples

Building blocks for FFF-MALS

The Eclipse separation system integrates with top-quality HPLC components, including pumps, autosamplers, detectors and fraction collectors, from some of the leading HPLC manufacturers.

Wyatt can configure your current HPLC instruments for FFF-MALS, or provide a complete turn-key solution with your choice of HPLC from Agilent, Thermo or Shimadzu. The Eclipse and HPLC modules can be operated from the VISION CSH® chromatography software hub or through control plug-ins available for Agilent OpenLAB®, Thermo Chromeleon® and Shimadzu LCSolutions®.

Add one or more of these detector modules to analyze molecular weight, size and conformation:

- DAWN®, the essential MALS detector for characterizing molar mass from 200 Da to 1 GDa and size (rms radius R_g) from 10 nm to 500 nm
- WyattQELS™ dynamic light scattering module, embeds inside the DAWN, for measuring size (hydrodynamic radius, R_h) from 0.5 to 300 nm
- Optilab®, the leading differential refractometer for universal concentration measurement
- ViscoStar®, the premier differential viscometer for characterizing polymer conformation and size

In addition, hydrodynamic size may be determined from AF4 retention time. VISION CSH collects, records and analyzes all the necessary signals for maximal sample characterization.



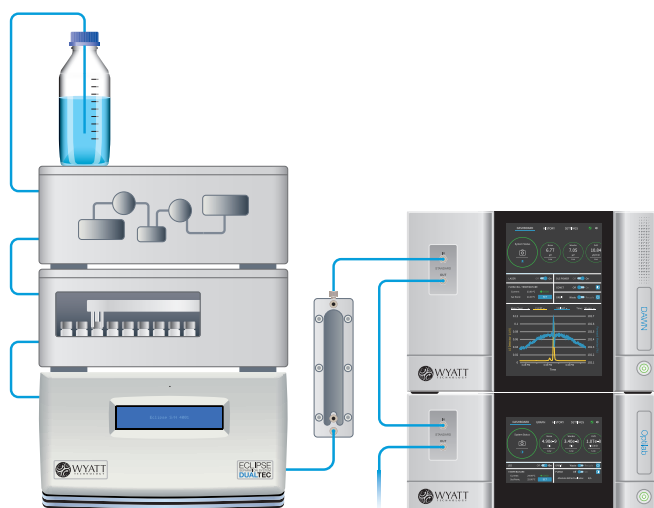
Eclipse DualTec

More automation and flexibility

Automated switching

FFF or SEC? The DualTec gives you the best of both worlds: It switches automatically between any two of SEC, AF4 and HF5, using the same HPLC and detector modules for each method.

- Compare results on SEC and HF5
- Develop a method on HF5 and transfer to AF4



Automated focusing

The DualTec's unique AutoFocus feature relieves the operator of manually focusing the sample, making for a seamless workflow from injection to analysis.

Metal-free option

Switch to a completely metal-free path at the push of a button. This configuration is optimal for FFF-ICP-MS, where metal ions can interfere with the analysis.

Additional benefits of the DualTec

- Tip-injection technology supports HF5 and requires only inlet and outlet ports for the separation channel. This simplifies the system relative to the traditional three-port configuration.
- Realtime measurement of channel flow and cross-flow rate achieved with a second Coriolis flowmeter. This allows direct verification of all separation parameters.

Eclipse AF4

Frit inlet and semi-preparative capabilities

The AF4 Standard

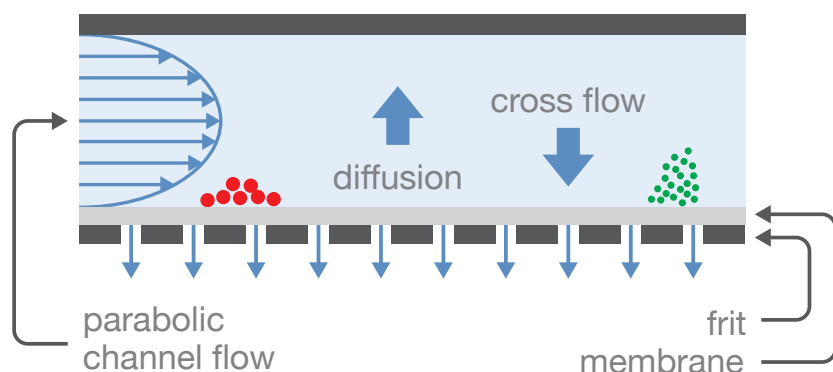
Users of the Eclipse AF4 system enjoy the benefits of the essential Eclipse approach to flow FFF: The unique single-pump design, which does away with the complication, reduced reliability and increased pulsation of a multiple-pump flow FFF system. The Eclipse AF4 features center-downstream injection and does not support tip-injection or HF5. However, for specific applications it offers specialized separation modes:

- Frit-inlet (FI) channel: The AF4 supports Wyatt's unique FI channel that eliminates the need to focus the sample, preventing aggregation of sticky samples.
- Semi-preparative (SP): Wyatt's unique SP channel can load milligrams of protein, up to 100x more than a typical analytical AF4 injection. Use with the HPLC's fraction collector to pool purified fractions for later use.



How Asymmetric-Flow FFF works...

Gentle and fast one-phase separation



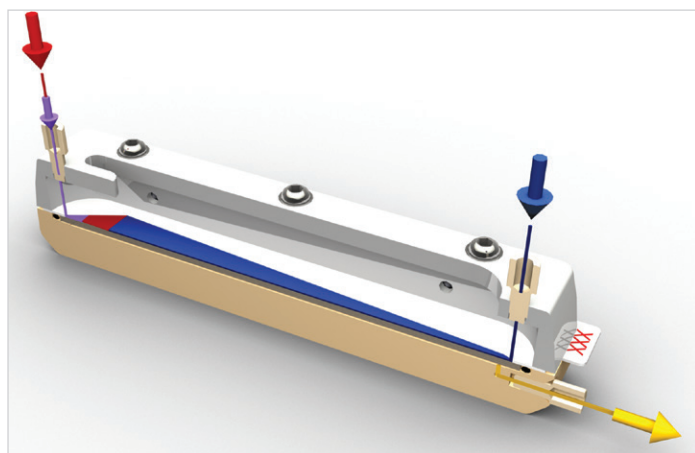
Asymmetric-Flow Field-Flow Fractionation (AF4) Flat Channel: FFF separation power can be tuned by changing the ratio of cross flow to channel flow.

In AF4, there is no stationary phase; the channel constitutes an open flow path between two parallel plates. The top plate is solid and the bottom plate is a semi-permeable membrane supported by a frit. While solvent can pass through the membrane, samples larger than the membrane cutoff size cannot. A flow restrictor on the channel outlet forces a portion of the flow, known as the cross flow, to pass through the membrane. Samples are introduced into the flow stream at the top of the separation channel (red arrow) and focused along a narrow band of the membrane by opposing channel flows (red and blue arrows) which pass through the semi-permeable membrane.

The cross-flow force pushing the sample against the membrane is balanced by diffusion, resulting in an equilibrium height distribution above the membrane that depends on particle size. Smaller particles are, on average, higher up in the channel compared to larger particles.

After establishing equilibrium height distributions, channel flow is applied parallel to the membrane,

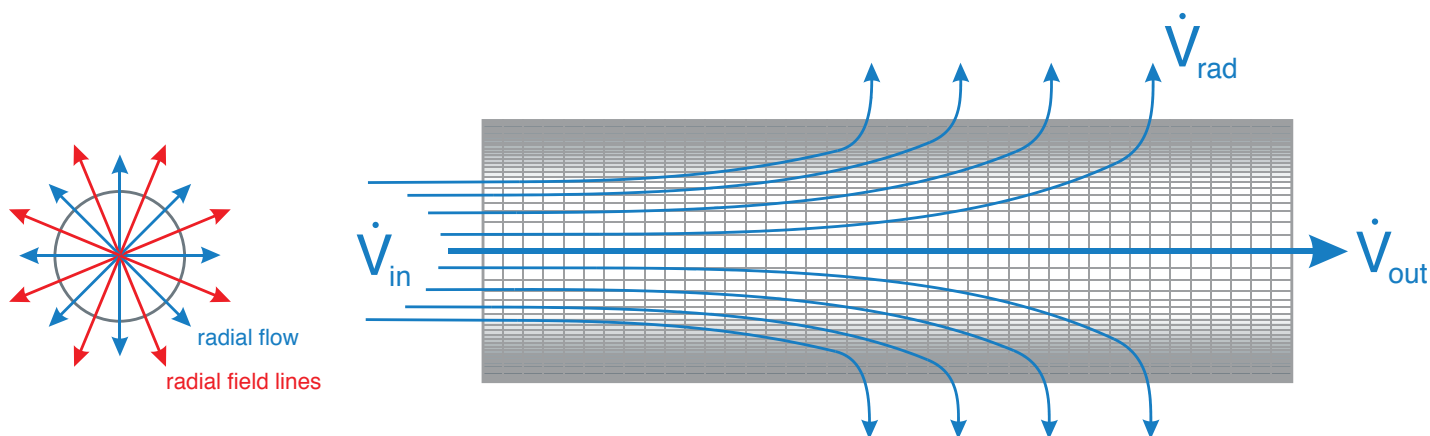
creating a laminar flow stream with a parabolic velocity profile. The flow velocity strongly increases with its distance from the wall. Hence smaller particles are transported more rapidly along the channel relative to larger particles. Finally, the particles are swept out of the channel, downstream to the detectors, eluting in order of increasing size.



The DualTec AF4 channel uses tip-injection. The sample is introduced with the flow entering the channel inlet prior to focusing.

...and Hollow-Fiber Flow FFF

Higher sensitivity and separation efficiency



HF5 presents a simple geometry:

The cross flow is radially symmetric. The separation channel consists of a semi-porous, hollow tube, in one continuous piece.

Hollow-Fiber Flow FFF (HF5) uses a tubular polymeric fiber with porous walls as the separation channel. Cross flow in the fiber is radial instead of unidirectional. These fibers allow a high flux, are mechanically stable and exhibit minimal interaction with most types of samples, particularly proteins.

HF5 offers three primary advantages over AF4:

- Disposable cartridges
- Higher sensitivity
- Higher separation efficiency (plate numbers)

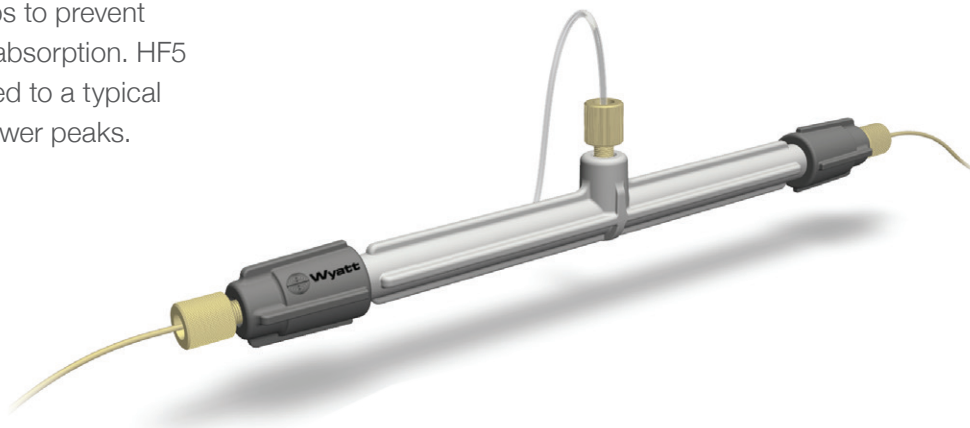
In HF5, the same retention results in 33% lower concentration levels compared to AF4. This helps to prevent overloading and sample aggregation or absorption. HF5 encapsulates 10x lower volume compared to a typical AF4 channel, with correspondingly narrower peaks.

Detector flow rates can also be reduced, leading to lower dilution factors. Together, narrow peaks and low dilution contribute to increased sensitivity. A side benefit of these is greatly reduced solvent consumption.

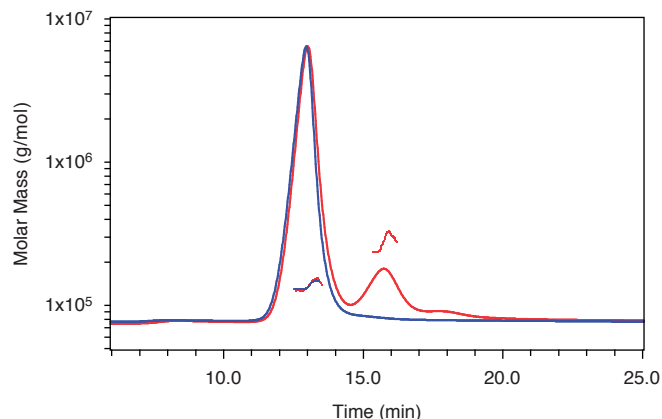
The disposable HF5 channel is a major break-through in FFF technology, as it greatly simplifies the effort of membrane replacement. Still, an HF5 cartridge will typically be usable over 100 injections.

The combination of long life, convenient and inexpensive replacement, reduction in sample and solvent consumption and faster run times adds up to a major productivity boost to FFF users.

The HF5 channel cartridge has a proprietary design and is intended to be disposable. Replacement of the channel simply requires reconnecting three finger-tight nuts.

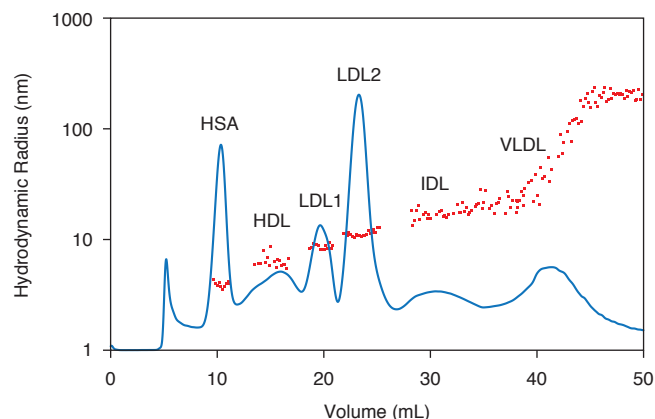


Monoclonal antibodies and aggregates



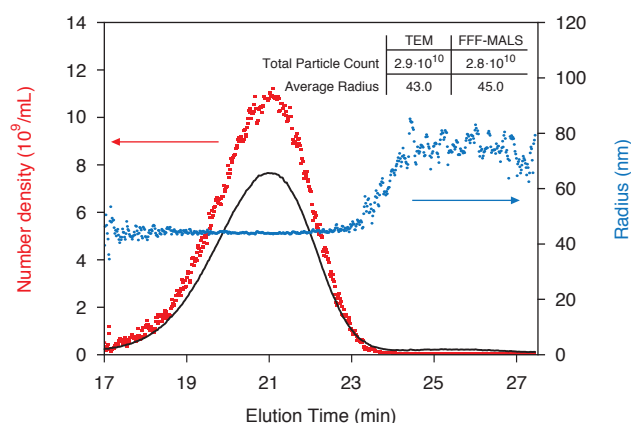
HF5-MALS analysis of two different preparations of a monoclonal antibody (mAb), using an Eclipse DualTec and an HF5 cartridge: I (UV trace in blue) and II (red). HF5 exhibits excellent resolution between monomer, dimer and higher oligomers, with high reproducibility. The injected mass was only 2 μ g.

Whole serum



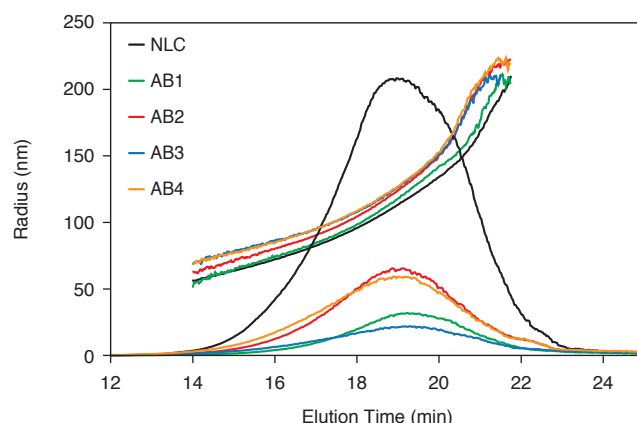
AF4-MALS-DLS separation of whole serum, with distinct peaks for serum albumin, and various types of lipoproteins. Sizes (R_h) were determined by online dynamic light scattering embedded in the DAWN MALS detector. Not shown, MALS also determines molar masses of each peak and, for species larger than ~ 10 nm, rms radius R_g .

Adenovirus concentration



AF4-MALS provides quantitative, high-resolution size distributions with large particle ensembles for statistical certainty. This adenovirus analysis indicates the number density in billion/mL at each elution time, along with the radius, both of which compare very well with TEM. The LS fractogram is overlaid in black. A small fraction of dimers is evident.

Silver-nanolipid complexes



AF4-MALS analysis of a nanolipid complex (NLC) and microsilver-loaded NLC formulations with varying concentrations of NLC and microsilver. All preps show a higher radius than the pure NLC, proving the adsorption of silver ions. The size saturated at 10% w/w NLC and 0.1% w/w Ag. Reproducibility is better than 1%.

Specifications

Separation Range	Approximately 1 nm to 10 μ m, depending on specific sample properties and the membrane installed
Control Unit	
Solvent Compatibility	<i>Standard Version:</i> For aqueous solvents and alcohols. Corrosion resistant from pH 2 to 12. Wetted materials: PEEK, stainless steel, Kalrez, Viton; <i>Organic Version:</i> For organic solvents such as THF, toluene, etc. Includes vapor sensor with interlock output. Wetted materials: PEEK, stainless steel, Kalrez; <i>Metal-Free Version (DualTec only):</i> Wetted materials: PEEK
Supported HPLC Software Control	Agilent, Thermo, Shimadzu Eclipse software plug-ins for Agilent OpenLAB for Chemstation, Thermo Chromeleon and Shimadzu LabSolution; VOYAGER CDS® stand-alone chromatography data system controls Eclipse, Agilent and Shimadzu HPLC modules and ASTRA® software to initiate detector data acquisition SCOUT DPS® for virtual method development, method optimization and particle size analysis
Additional Software	Ethernet
Communications Inputs/Outputs	Autoinject In, Autoinject Out, Analog In, Pump Analog Control Out, Detector Zero Out, Start Pump, Pump Interlock for Agilent pumps
Dimensions	64 cm (L) x 36 cm (W) x 25 cm (H)
AF4 Channels Selection	<i>Analytical AF4 channels:</i> short, 153 mm effective length (e.l.); long, 246 mm e.l., frit-inlet, 246 mm e.l. Each analytical channel is available in aqueous and organic versions <i>Semi-prep channel:</i> (Eclipse AF4 model only), aqueous version only <i>Aqueous channels:</i> Stainless steel, PEEK, polycarbonate, Viton <i>Organic channels:</i> Stainless steel, PEEK, Kalrez
Wetted Materials	Maximum 30 bar, with software overpressure protection Spacer thicknesses are 250, 350 and 490 μ m
Channel Pressure Spacers	Pre-cut: Polyether Sulfone: 1, 5, 10, and 30 kDa cutoff; Regenerated cellulose: 10 and 30 kDa cutoff. Other membranes may be cut from sheets with the Membrane Template Kit.
Membranes	
Hollow-Fiber Cartridge Selection	PES membrane 10 kDa cutoff, channel volume 90 μ l, e.l. 170 mm, channel height 400 μ m. Aqueous solvents only.
Dimensions	64 cm (L) x 36 cm (W) x 25 cm (H)
Weight	20 kg

Wyatt Technology is committed to continual improvement. Specifications subject to change without notice.

Warranty: All Wyatt instruments are guaranteed against manufacturing defects for 1 year.

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Dr. Philip J. Wyatt, Chairman of the Board
Clifford D. Wyatt, President

For more than 35 years, we've operated as one of the very few remaining family-owned businesses in the analytical instrument industry. With installations in more than 65 countries, over 15,000 refereed journal publications citing our instruments and more than 25 PhD scientists on staff, we take great pride in the worldwide recognition that Wyatt Technology has received as a leading manufacturer of instruments and software for absolute macromolecular and nanoparticle characterization. Our dedication to providing customers with comprehensive training and personal support has made us the gold standard in this field.

The Eclipse is one of many tools in Wyatt's Light Scattering Toolkit for Essential Macromolecular and Nanoparticle Characterization.

Learn more at www.wyatt.com